NM4QM 2026 · HANDS-ON SESSION · 45 MIN

Spin and anomalous Hall effects with WannierBerri and GPAW

Symmetry-adapted Wannier functions, spin–orbit coupling on demand, the spin Hall effect of graphene and the anomalous Hall effect of the altermagnet MnTe

Yaroslav Zhumagulov
Chair of Computational Physics of Condensed Matter (C3MP), EPFL, Lausanne, Switzerland
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THE STORY

Two puzzles, two acts

0–8 min	8–19	19–37	37–45
Setup and ideas	Act 1: graphene	Act 2: MnTe	Your material, questions

TWO PUZZLES

- Spin–orbit coupling turns graphene into a quantum spin Hall insulator with a gap of 42 μeV
- MnTe has no net magnetisation, yet shows an anomalous Hall effect

WHAT YOU WILL DO

- Build symmetry-adapted Wannier functions straight from GPAW
- Switch spin–orbit coupling on, off and around, without new DFT
- Compute spin and anomalous Hall conductivities

Kane, Mele, PRL 95, 226801 (2005) · Gonzalez Betancourt et al., PRL 130, 036702 (2023)

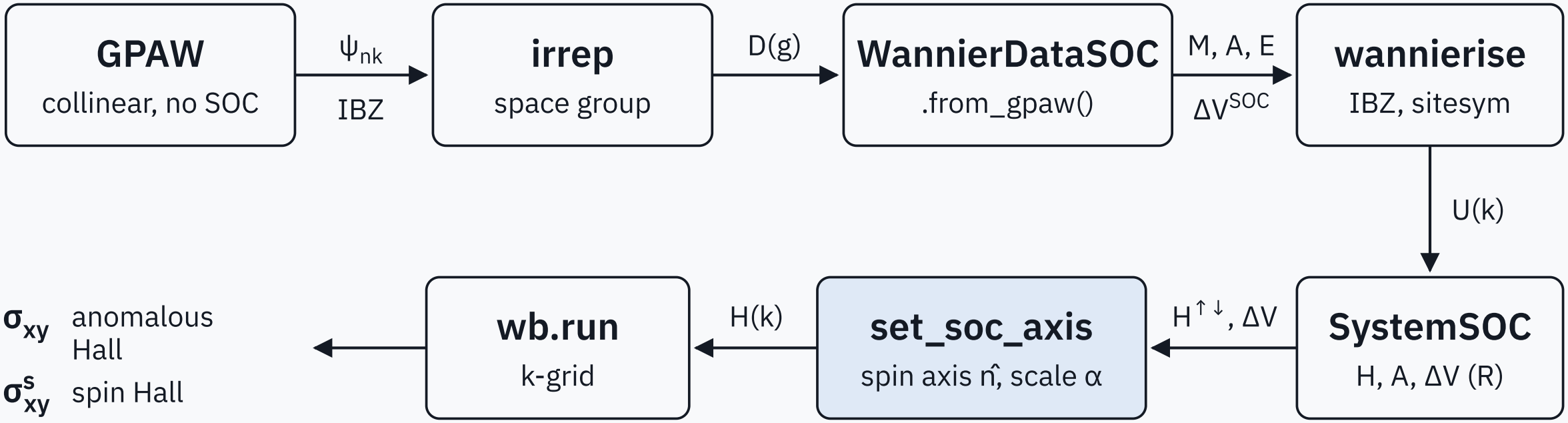
SETUP

Before we start: one check, 20 seconds

```
cd wannier-berri-demo && conda activate wbdemo
python setup/check_setup.py        # should end with "Ready."
cd 01_graphene
python 1_dft.py                    # 10 s
jupyter lab graphene.ipynb
```

- No environment yet: `bash setup/setup.sh` (15 min)
- DFT fails: `tar -xzf dft-files.tar.gz` in `wannier-berri-demo/`
- Fell behind: `bash run_all.sh` (6 min); finished notebooks in `reference/`
- Numbers on the slides are checkpoints: compare them with yours

One Python pipeline, GPAW to conductivities



Each box is one call in the notebooks. The shaded step needs no new DFT and no new Wannierisation.

$\psi_{n\mathbf{k}}$: Bloch states · $D(g)$: symmetry matrices on them · M, A, E: overlaps, projections, energies · ΔV^{SOC}: SOC matrix elements from the PAW spheres · $U(\mathbf{k})$: Wannier gauge · H, A, $\Delta V(\mathbf{R})$: Hamiltonian, Berry connection and SOC in the Wannier basis · $\hat{\mathbf{n}}$, α: spin axis, SOC scale · σ_{xy}: anomalous Hall (charge current), σ^{s}_{xy}: spin Hall (spin current) conductivity

Berry curvature needs millions of k-points; DFT gives tens

Intrinsic anomalous Hall conductivity:

$$\sigma_{xy} = -\frac{e^2}{\hbar} \int_{\mathrm{BZ}} \frac{d^3k}{(2\pi)^3} \sum_n f(\varepsilon_{n\mathbf{k}})\, \Omega_n^z(\mathbf{k})$$

$$\Omega_n^z = -2\,\mathrm{Im} \sum_{m \neq n} \frac{v_{nm}^x\, v_{mn}^y}{(\varepsilon_n - \varepsilon_m)^2}$$

f: occupation, ε_n: band energy, $v_{nm}^a = \langle u_n | \partial_{k_a} H | u_m \rangle$. Ω_n^z is large and narrow wherever two bands come close.

Wannier functions give $H(\mathbf{k})$ at any $\mathbf{k}$:

$$H_{ij}(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{R}}\, \langle w_{i\mathbf{0}} | \hat{H} | w_{j\mathbf{R}} \rangle$$

$w_{j\mathbf{R}}$: Wannier function j in the cell at $\mathbf{R}$; a few hundred $\mathbf{R}$ are enough.

7 → 1.44 M
graphene: DFT k-points → spin Hall grid

24 → 0.59 M
MnTe: DFT k-points → anomalous Hall grid

30 s, 1 min
the two integrals on an 8-core laptop

Xiao, Chang, Niu, RMP 82, 1959 (2010) · Wang et al., PRB 74, 195118 (2006) · Marzari et al., RMP 84, 1419 (2012)

Symmetry-adapted Wannier functions: DFT on irreducible k only

Each Wannier function should transform like its orbital:

$$\hat{g}\, w_{i\mathbf{R}}(\mathbf{r}) = \sum_j D_{ji}(g)\, w_{j,\, g\mathbf{R}}(\mathbf{r})$$

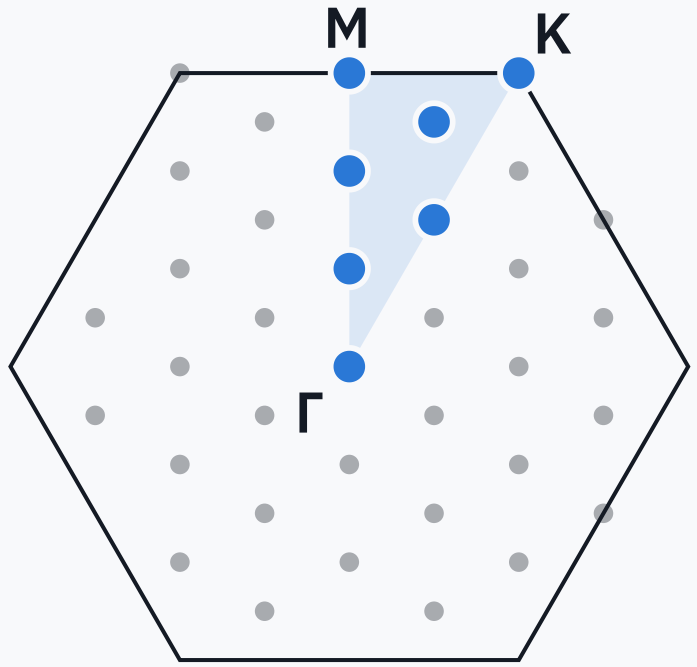
$\hat{g}$: symmetry operation, $D(g)$: its matrix on the orbitals of one site

Then the gauge at g**k** follows from **k** (Sakuma 2013):

$$U(g\mathbf{k}) = \tilde{d}(g,\mathbf{k})\, U(\mathbf{k})\, D(g,\mathbf{k})^{\dagger}$$

U: Bloch → Wannier gauge, $\tilde{d}$: g acting on the Bloch states (from irrep)

- DFT on 7 of 36 k-points for graphene, 24 of 144 for MnTe
- Degeneracies come out exact



6 × 6 grid: 36 points, 7 irreducible (blue)

METHODS

Wannierisation: WannierBerri vs Wannier90

$$\Omega = \sum_n \left[\langle w_n | r^2 | w_n \rangle - \left| \langle w_n | \mathbf{r} | w_n \rangle \right|^2 \right] = \Omega_{\mathrm{I}} + \tilde{\Omega}$$

Ω: spread of the Wannier functions · Ω_{I}: fixed by the chosen subspace (disentanglement) · $\tilde{\Omega}$: depends on the gauge $U(\mathbf{k})$ (localisation)

	Wannier90	**WannierBerri, this demo**
Input	.amn, .mmn, .eig files from a DFT interface	computed in Python from the GPAW wave functions
k-points	full grid; irreducible ones with site_symmetry	irreducible ones
Algorithm	disentangle first, then minimise $\tilde{\Omega}$ by steepest descent or conjugate gradients	one loop: a disentanglement step and a local gauge update per iteration, no gradient descent
Symmetry	site_symmetry with a .dmn file from pw2wannier90, not with a frozen window	from irrep, for any code it reads; frozen window allowed
SOC	spinor DFT run with SOC	collinear run, SOC added afterwards; spin ↓ from symmetry in altermagnets

Marzari, Vanderbilt, PRB 56, 12847 (1997) · Souza, Marzari, Vanderbilt, PRB 65, 035109 (2001) · Sakuma, PRB 87, 235109 (2013) · Pizzi et al., JPCM 32, 165902 (2020)

METHODS

Post-processing: WannierBerri vs postw90

$$H(\mathbf{K}+\boldsymbol{\kappa}) = \sum_{\mathbf{R}} e^{i\boldsymbol{\kappa}\cdot\mathbf{R}} \left[e^{i\mathbf{K}\cdot\mathbf{R}}\, H(\mathbf{R})\right]$$

$\boldsymbol{\kappa}$: points of a coarse FFT grid · $\mathbf{K}$: shifts of that grid; one FFT per shift, the shifts together give the dense grid

	postw90	**WannierBerri**
$H(\mathbf{R}) \to H(\mathbf{k})$	sum over **R** at every k, cost $\propto N_{\mathbf{R}}$	FFT over sub-grids, cost $\propto \log N_{\mathrm{FFT}}$ per k
Symmetry	full grid	irreducible points of the magnetic point group, result symmetrised
Hot spots	finer mesh where |Ω| exceeds a threshold	recursive adaptive refinement
Fermi levels	cost grows with their number	~1000 at almost no extra cost
SOC axis	fixed by the DFT run	`set_soc_axis` : any axis and strength

Up to three orders of magnitude faster on dense grids. In this demo: 1401 Fermi levels in one run; for $\hat{\mathbf{n}} \parallel y$ only the k-points irreducible under m′m′m (8 operations) are computed.

Wang, Yates, Souza, Vanderbilt, PRB 74, 195118 (2006) · Tsirkin, npj Comput. Mater. 7, 33 (2021) · Pizzi et al., JPCM 32, 165902 (2020)

Does graphene have a gap?

Step	Run	Checkpoint
0 · DFT	`python 1_dft.py`	10 s, `gpaw/nscf.gpw`
1 · Symmetry	irrep	48 operations, 7 of 36 k-points
2 · Wannier functions	`wannierise`	two equal p_z-like functions
3 · Bands	interpolation	blue on grey in the window
4 · SOC on demand	`set_soc_axis`	gap ≈ 36 μeV (GPAW ≈ 39 μeV)
5 · Spin Hall	`wb.run`	plateau at e/2π, 30 s

Graphene: DFT in 10 s, two p_z Wannier functions

- PBE, 400 eV, 6×6×1 grid, 15 Å vacuum, no SOC
- NSCF only on the 7 irreducible k-points
- Frozen window E_F − 2.9 … E_F + 2.8 eV, plus the π state at Γ

2
equal, compact p_z functions

48
symmetry operations

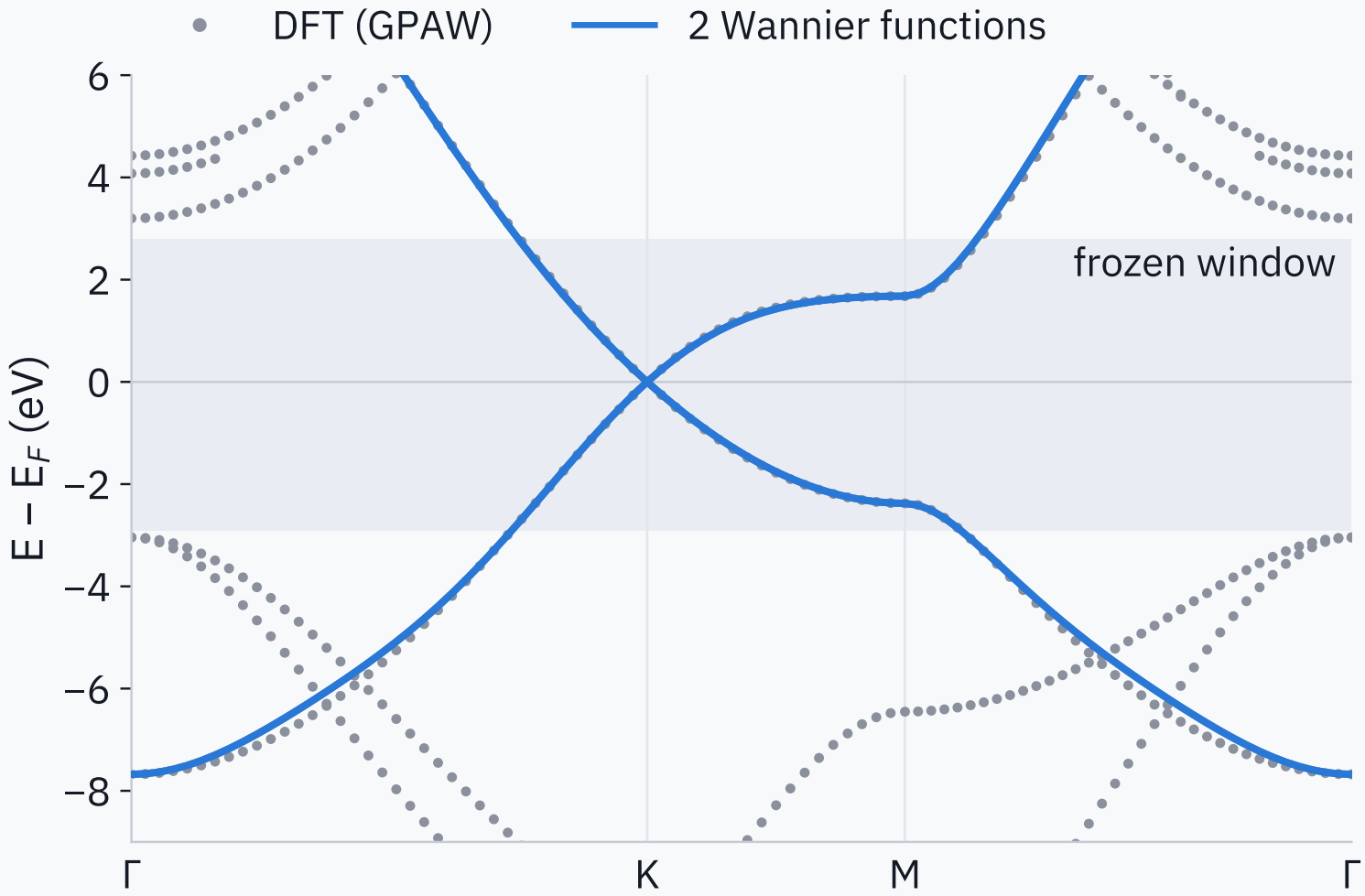
3 s + 2 s
from_gpaw + wannierise

```
calc = GPAW(mode=PW(400), xc="PBE",
            kpts={"size": (6, 6, 1), "gamma": True})
atoms.calc = calc
atoms.get_potential_energy()
sg = SpaceGroup.from_gpaw(calc)                    # irrep
kpts = sg.get_irreducible_kpoints_grid((6, 6, 1))  # 7 of 36
calc.fixed_density(kpts=kpts, nbands=20).write(
    "gpaw/nscf.gpw", mode="all")

# graphene.ipynb
pz = Projection(position_num=[[1/3, 2/3, 0], [2/3, 1/3, 0]],
                orbital="pz", spacegroup=sg)
wandata = wb.WannierDataSOC.from_gpaw(
    calc, projections=ProjectionsSet([pz]))
wandata.wannierise(froz_min=EF - 2.9, froz_max=EF + 2.8,
                   frozen_states={0: [1]},  # pi at Gamma
                   num_iter=100, sitesym=True)
system = wb.SystemSOC.from_wannierdata(wandata, berry=True)
```

Two Wannier bands follow DFT through the frozen window

- Grey: GPAW along Γ–K–M–Γ, not used in the fit
- Blue: the two-band Wannier model
- Shaded: frozen window



SOC as a matrix you can rotate and scale

$$H(\mathbf{k}) = H_0(\mathbf{k}) \otimes \sigma_0 + \alpha \sum_c \Delta V^c(\mathbf{k}) \otimes \sigma_c(\hat{\mathbf{n}})$$

$$\Delta V^c_{mn}(\mathbf{k}) = \sum_{a,ij} \langle \tilde{\psi}_{m\mathbf{k}} | \tilde{p}^a_i \rangle \, \langle \phi^a_i | \, \xi_a(r) \, L_c \, | \phi^a_j \rangle \, \langle \tilde{p}^a_j | \tilde{\psi}_{n\mathbf{k}} \rangle$$

$$\xi_a(r) = \frac{1}{4m^2c^2} \frac{1}{r} \frac{dV_a}{dr}$$

H_0: Wannier Hamiltonian without SOC · $\sigma_c(\hat{\mathbf{n}})$: Pauli matrices, z along the spin axis $\hat{\mathbf{n}}$ · α: SOC scale, 1 = real · $\tilde{\psi}, \tilde{p}^a_i, \phi^a_i$: PAW pseudo wave functions, projectors, partial waves of atom a · L_c: orbital angular momentum · V_a: potential in sphere a

- Same PAW matrix elements as GPAW's own SOC
- New $\hat{\mathbf{n}}$ or α: no new DFT

```
K = [1/3, 1/3, 0]
energy = wb.calculators.tabulate.Energy(
    degen_thresh=1e-8)
for alpha in [0, 1]:
    system.set_soc_axis(alpha_soc=alpha)
    E = wb.evaluate_k(system, k=K,
            calculators={"E": energy})
```

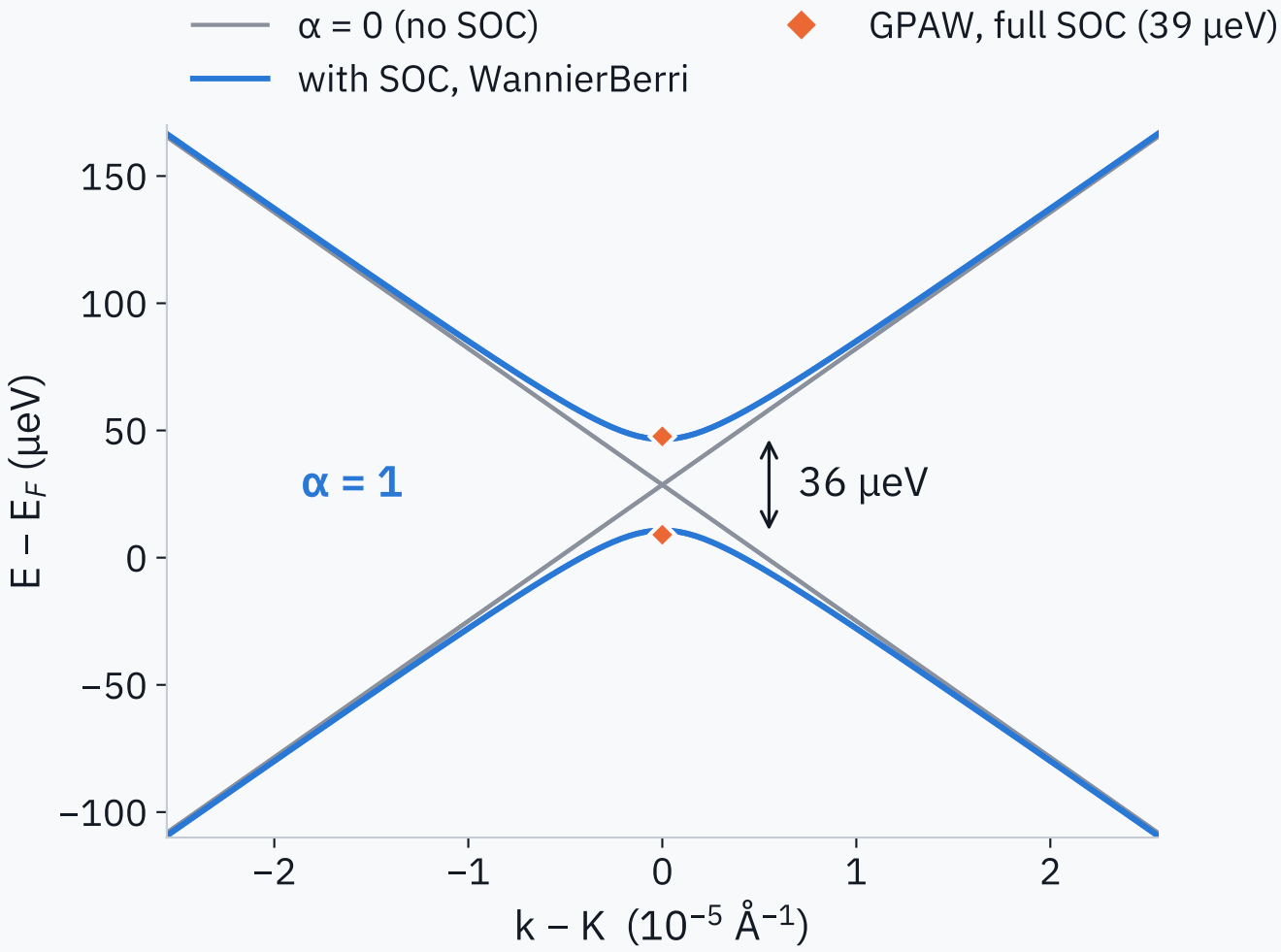
```
             Dirac point     gap at K
alpha = 0    +29 ueV          0
alpha = 1    +29 ueV         36 ueV
GPAW SOC     +28 ueV         39 ueV
```

`degen_thresh=1e-8` : the default 10^{-4} eV would merge the split pair

Kane–Mele gap: 36 µeV from two Wannier functions

Gap at K: **≈ 36 µeV**
GPAW, full SOC, 20 bands: **≈ 39 µeV**

- p_z has $m = 0$, so the gap comes from the small d part of p_z in the PAW spheres (Gmitra 2009)
- Missing 3 µeV: second-order SOC via the σ bands
- Depends on cutoff and PAW dataset; experiment 42 µeV (Sichau 2019)

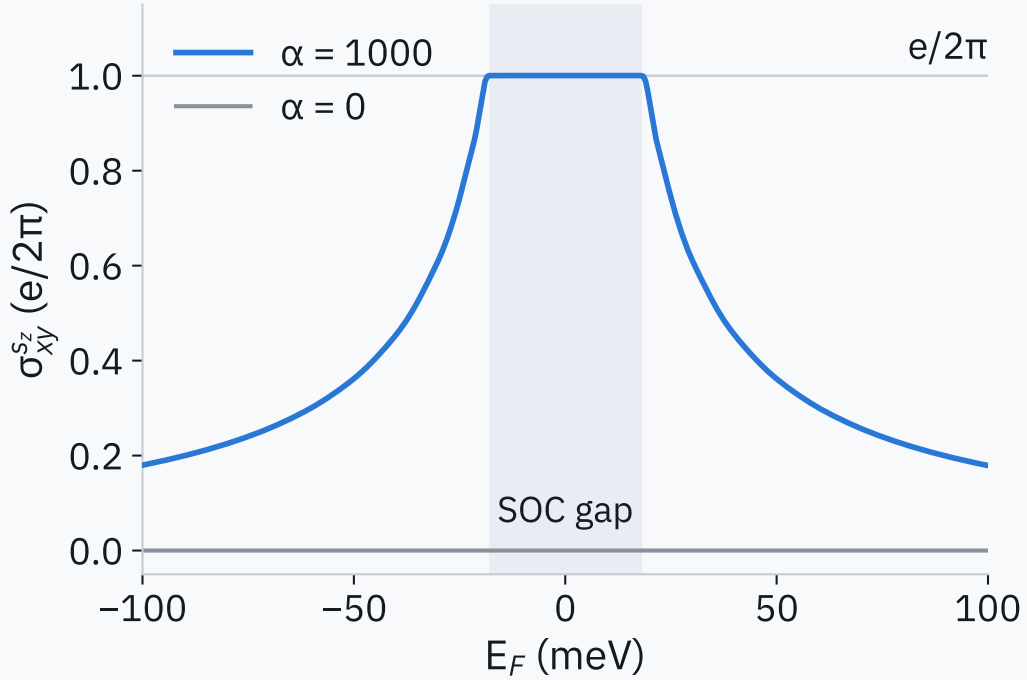


Kane, Mele, PRL 95, 226801 (2005) · Gmitra et al., PRB 80, 235431 (2009) · Sichau et al., PRL 122, 046403 (2019)

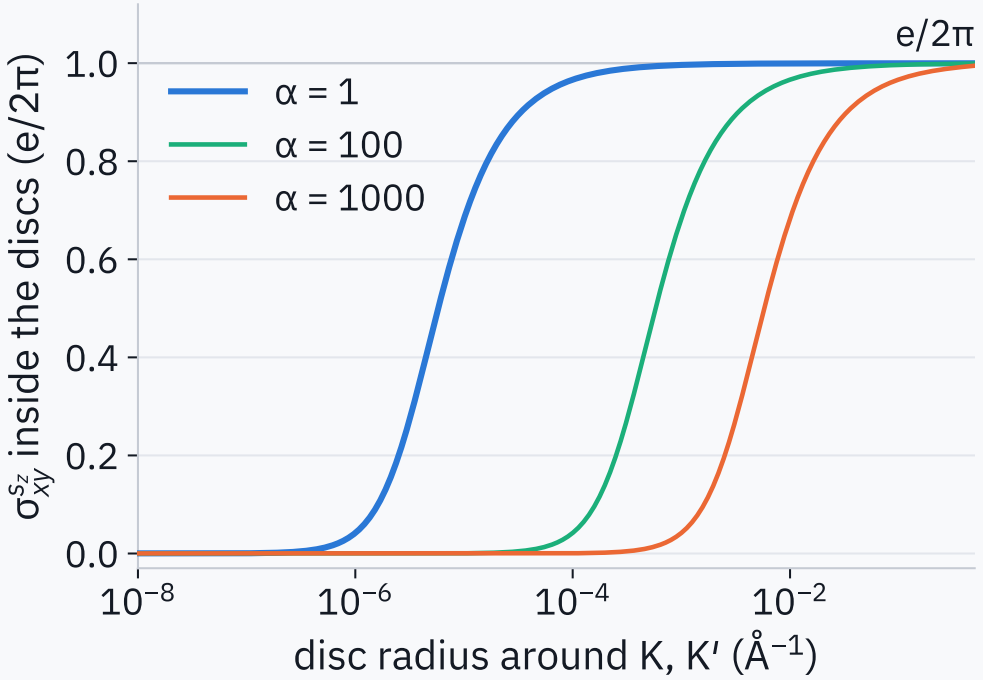
Spin Hall conductivity: e/2π, also at the real gap

$$\sigma_{xy}^{s_z} = \frac{e}{\hbar}\int \frac{d^2k}{(2\pi)^2}\sum_n f_{n\mathbf{k}}\,\Omega_n^{s_z}, \qquad \Omega_n^{s_z} = -2\hbar^2\,\mathrm{Im}\sum_{m\neq n}\frac{(j_x^{s_z})_{nm}(v_y)_{mn}}{(\varepsilon_n-\varepsilon_m)^2}$$

$j_x^{s_z} = \frac{1}{2}\{s_z, v_x\}$: spin current, v: velocity. With s_z conserved, $\sigma_{xy}^{s_z} = (e/2\pi)(C_\uparrow - C_\downarrow)/2 = e/2\pi$



α = 1000 (gap 36 meV): plateau at e/2π



α = 1, the real 36 μeV gap: also e/2π

ACT 2 · MNTE.IPYNB

A Hall effect without magnetisation

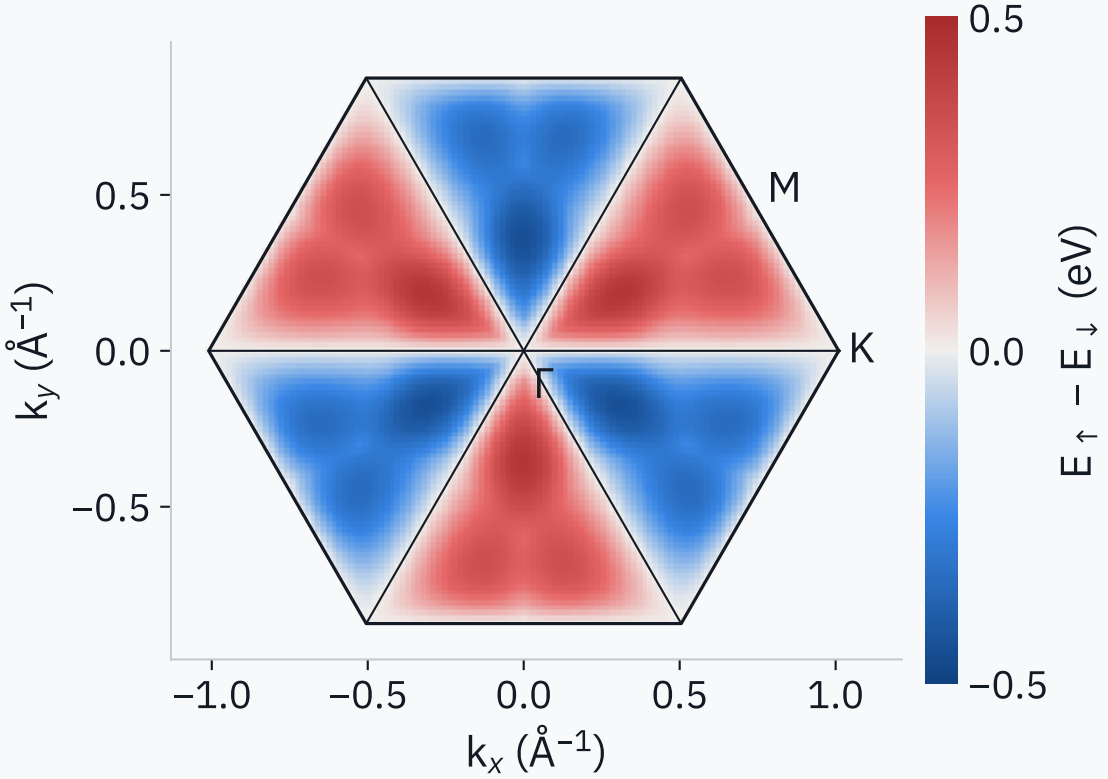
Step	Run	Checkpoint
0 · DFT	`python 1_dft.py`	45 s, moments ≈ ±4.4 μ_B
1–2 · Spin ↑ Wannier	`altermagnetic=True`	24 of 144 k-points, 25 s
3 · Splitting	no SOC	up to ≈ 1 eV at $k_z \neq 0$
4 · Turn $\hat{\mathbf{n}}$	`set_soc_axis`	SOC shift at A 4–5× smaller in plane
5 · Anomalous Hall	three $\hat{\mathbf{n}}$	0 for x and c, up to ≈ 80 S/cm for y
6 · Rotate in plane	0–30°	far from sin 3φ

Altermagnets: zero net moment, spin-split bands without SOC

- Two sublattices related by a rotation or mirror, not by a translation or inversion
- Bands spin split without SOC; the sign alternates in **k** (d-, g-, i-wave)
- No PT or *t*T symmetry, so with SOC an anomalous Hall effect is possible, depending on the Néel vector
- **MnTe**: NiAs type, $T_N \approx 310$ K, gap ≈ 1.3 eV

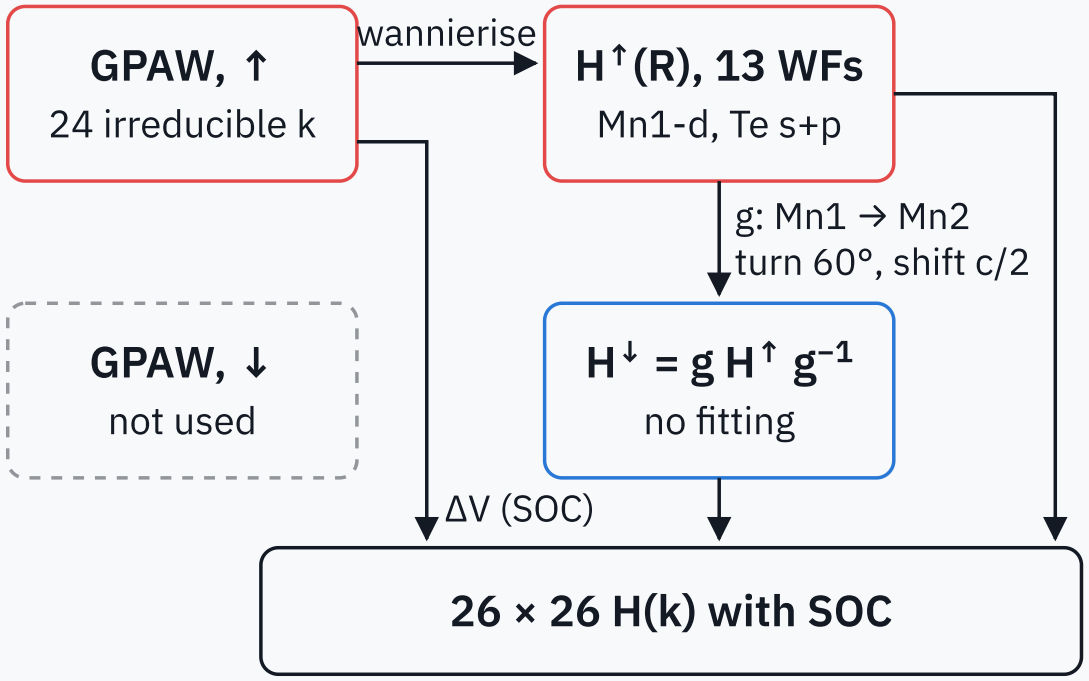
$$[C_2 \parallel g]: \qquad E_\uparrow(\mathbf{k}) = E_\downarrow(R_g\mathbf{k})$$

C_2: flips all spins, g: swaps the Mn sublattices, R_g: its rotation part



MnTe, no SOC: $E_\uparrow - E_\downarrow$ at $k_z = 0.35$ Å^{-1}

Wannierise spin ↑ only, spin ↓ follows by symmetry

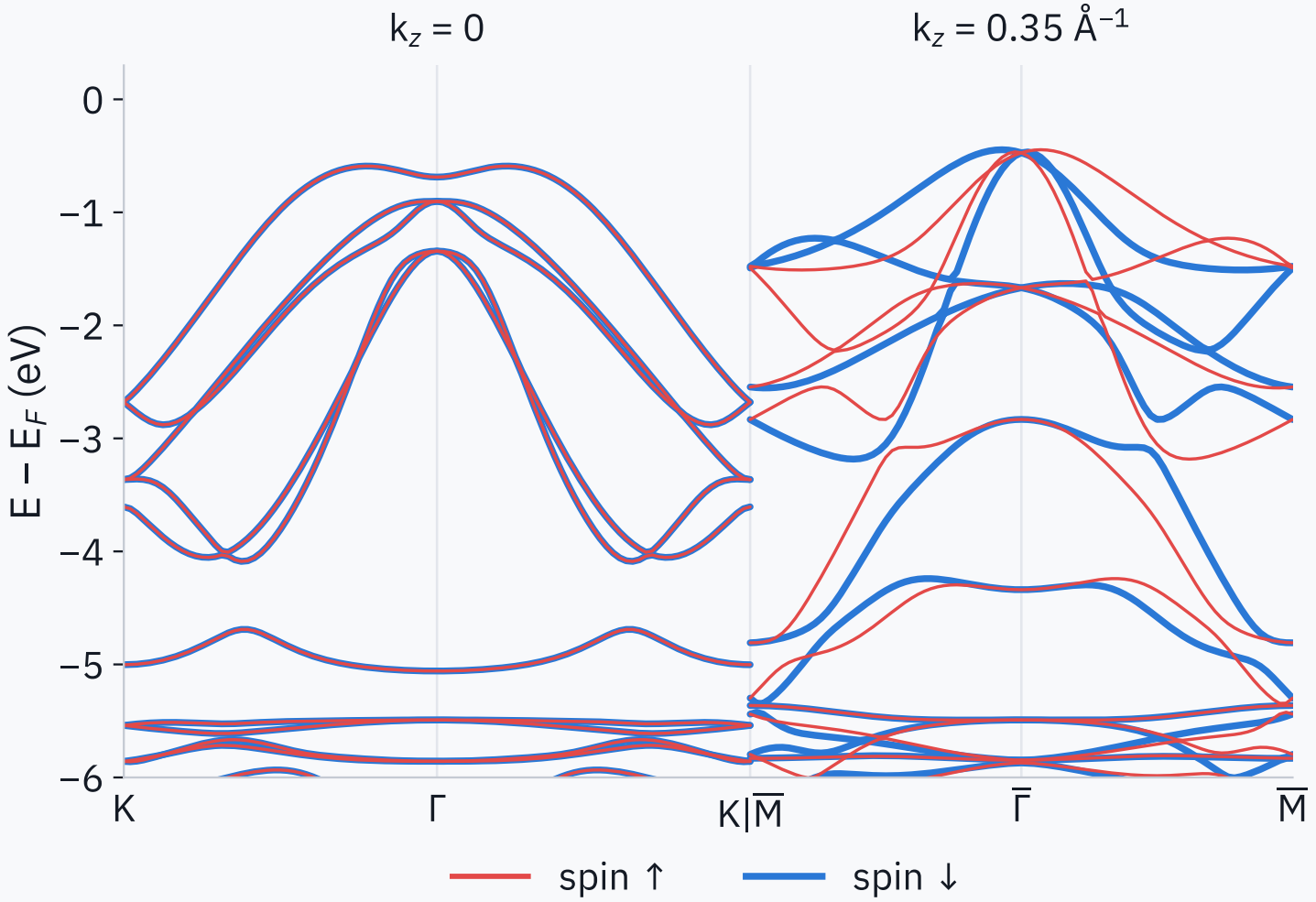


PBE+U (U = 4 eV): moments ≈ ±4.4 μ_B, gap ≈ 0.8 eV (exp. 1.3–1.4 eV; a larger U does not fix it)

```
calc = GPAW("gpaw/nscf.gpw")   # PBE+U, 24 of 144 k
sg = SpaceGroup.from_gpaw(calc)
Te = [[1/3, 2/3, 1/4], [2/3, 1/3, 3/4]]
projections = ProjectionsSet([
    Projection(position_num=[0, 0, 0], orbital="d",
               spacegroup=sg),          # Mn1 only
    Projection(position_num=Te, orbital="sp2",
               spacegroup=sg, xaxis=[0, -1, 0],
               rotate_basis=True),
    Projection(position_num=Te, orbital="pz",
               spacegroup=sg)])
wandata = wb.WannierDataSOC.from_gpaw(
    calc, projections=projections,
    altermagnetic=True,   # spin down from symmetry
    IBstart=8)            # skip Mn 3s, 3p
wandata.wannierise(froz_min=-np.inf,
                   froz_max=EF + 0.1,
                   num_iter=200, sitesym=True)
system = wb.SystemSOC.from_wannierdata(wandata,
                                       berry=True)
```

No SOC: degenerate at $k_z = 0$, split by up to 1 eV at $k_z \neq 0$

- Red: spin ↑, Wannierised
- Blue: spin ↓, from symmetry
- At $k_z = 0.35$ Å^{-1} up to ≈ 1 eV apart, with opposite order on the two sides of $\bar{\Gamma}$: g-wave



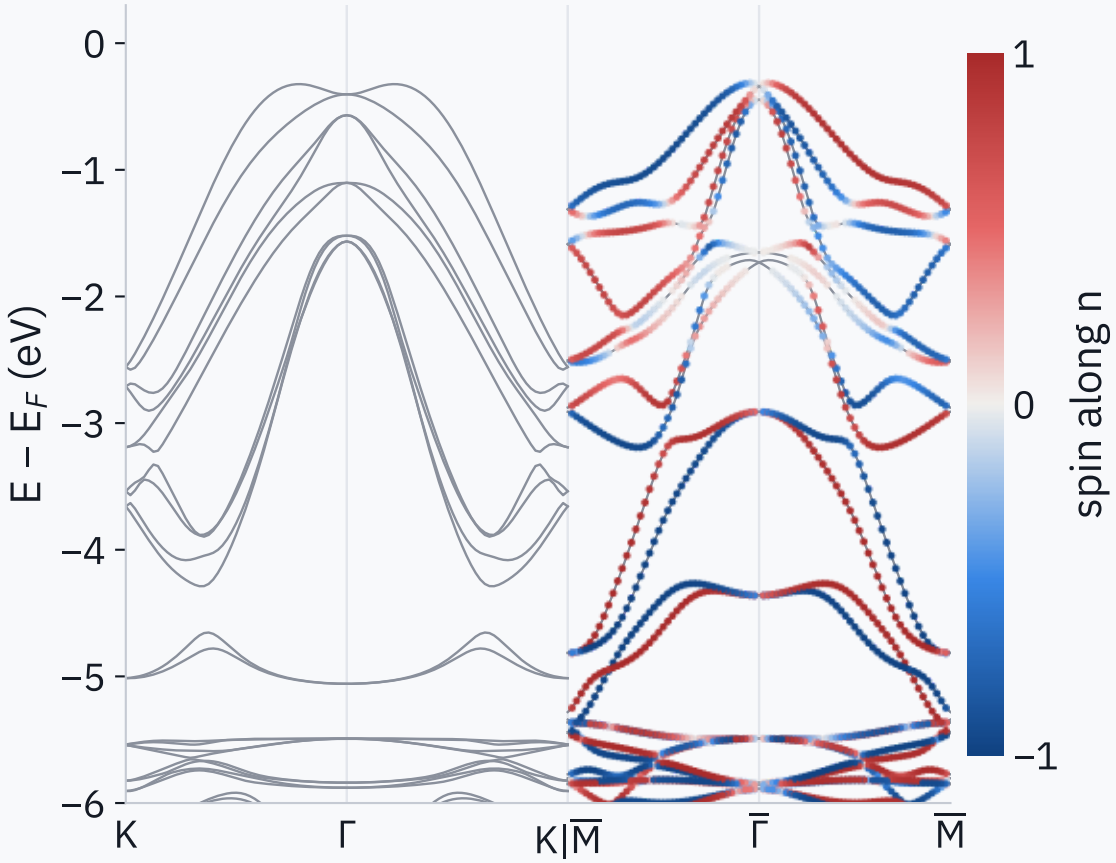
With SOC the Néel vector picks the magnetic point group

Néel vector n̂	Group	Hall vector
∥ c	6′/m′m′m	none
∥ a = x	mmm	none
⊥ a = y	m′m′m	$\sigma_{xy} \neq 0$

$$g: \ \boldsymbol{\sigma}_{\mathrm{H}} \to \det(R_g)\, R_g\, \boldsymbol{\sigma}_{\mathrm{H}}$$
$$\mathcal{T}: \ \boldsymbol{\sigma}_{\mathrm{H}} \to -\boldsymbol{\sigma}_{\mathrm{H}}$$

$\boldsymbol{\sigma}_{\mathrm{H}} = (\sigma_{yz}, \sigma_{zx}, \sigma_{xy})$: Hall vector · R_g: rotation part of g · $\mathcal{T}$: time reversal · primed: combined with $\mathcal{T}$

SOC shift of the top level at A: ≈ 0.3 eV for **n̂** ∥ c, 4–5 times smaller for **n̂** in plane; the model agrees with GPAW's full SOC

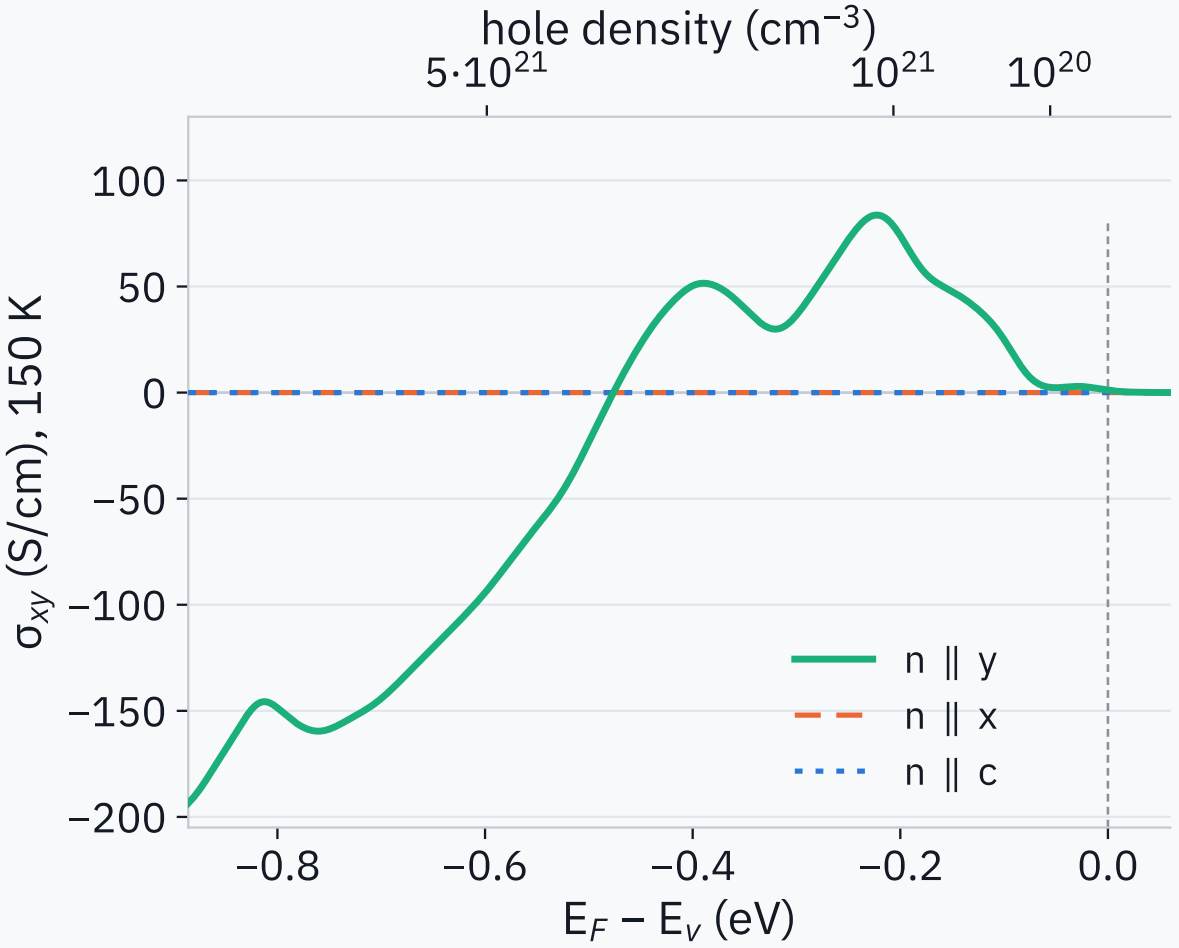


n̂ ∥ y, colour: spin along **n̂**

Anomalous Hall conductivity for three Néel vectors

```
ahc = wb.calculators.static.AHC(Efermi=Efermi,
    degen_thresh=1e-4, smoother=smoother)
directions = {"x": (90, 0), "y": (90, 90),
              "z": (0, 0)}   # theta, phi
grids = {"x": (48, 48, 32), "y": (96, 96, 64),
         "z": (48, 48, 32)}
for name, (theta, phi) in directions.items():
    system.set_soc_axis(theta=theta, phi=phi,
                        units="degrees")
    result = wb.run(system,
        grid=wb.Grid(system, NK=grids[name]),
        calculators={"ahc": ahc})
```

- $\hat{\mathbf{n}} \parallel x$ and $\hat{\mathbf{n}} \parallel c$: zero, as symmetry says
- $\hat{\mathbf{n}} \parallel y$: up to ≈ 80 S/cm, changes sign ≈ 0.5 eV below the valence band top E_V



Symmetry gives sin 3φ

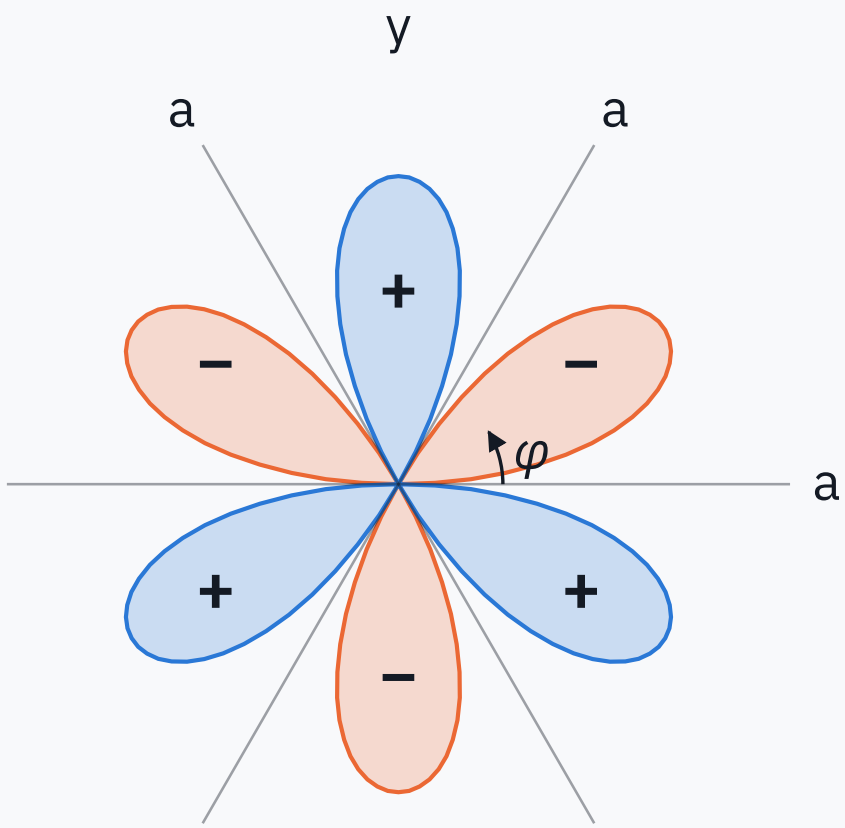
In-plane Néel vector at angle φ from a:

Operation	Swaps Mn	φ →	So
turn 180° about a	no	−φ	σ(−φ) = −σ(φ)
turn 60° about c, shift c/2, then $\mathcal{T}$	yes	φ + 60°	σ(φ + 60°) = −σ(φ)

$$\sigma_{xy}(\varphi) = A_3 \sin 3\varphi + A_9 \sin 9\varphi + A_{15} \sin 15\varphi + \ldots$$

A_m: amplitudes in S/cm

- Fixed by symmetry: zeros along every a axis, sign change every 60°
- Not fixed: the A_m; sin 3φ need not dominate



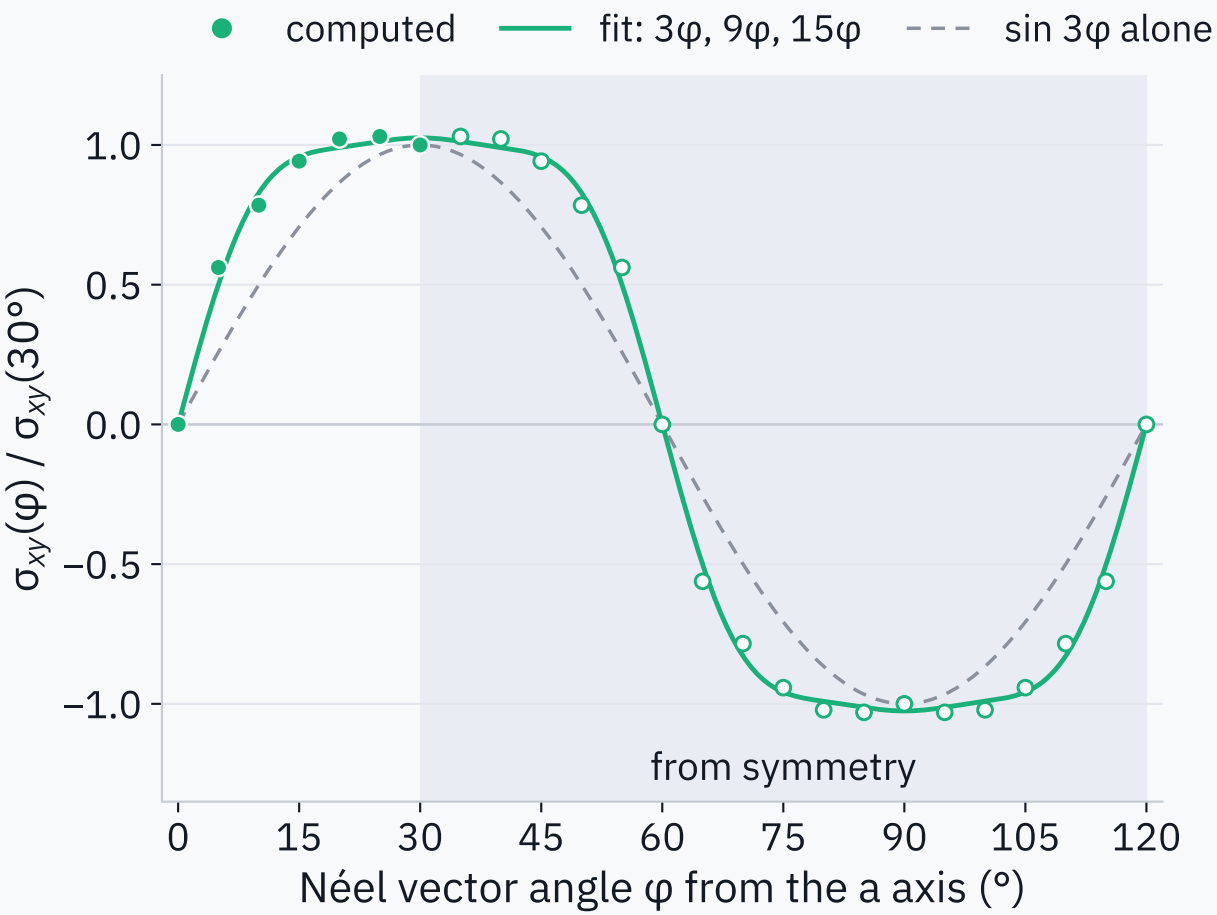
Sign of σ_{xy} against the direction of **n̂**, sin 3φ only (sketch)

Rotating the Néel vector: exact zeros, but not a sin 3φ curve

One `set_soc_axis` per angle, no new DFT: 0–30° in 5° steps, 1 min

- σ(−φ) = −σ(φ) and σ(φ + 60°) = −σ(φ) hold exactly
- Rises much faster than sin 3φ: σ(5°) is already half of σ(30°)
- σ ≈ A_3 (sin 3φ + 0.18 sin 9φ + 0.04 sin 15φ)

Strong hole doping: E_F 0.7 eV below the band top



Filled: computed. Open: from the two symmetry relations

Your material next

```
calc = GPAW(...)                                  # SCF
kpts = SpaceGroup.from_gpaw(calc)\
           .get_irreducible_kpoints_grid(grid)
calc.fixed_density(kpts=kpts, nbands=...)  # NSCF
wandata = wb.WannierDataSOC.from_gpaw(
    calc, projections=...)
wandata.wannierise(froz_min=..., froz_max=...,
                   sitesym=True)
system = wb.SystemSOC.from_wannierdata(
    wandata, berry=True)
system.set_soc_axis(theta, phi)
wb.run(system, grid=..., calculators=...)
```

- Projections: the orbitals that make the bands you care about
- Frozen window: every state you need kept exactly
- k-grids: multiples of the DFT grid
- `degen_thresh` below the smallest splitting
- Check a few levels against GPAW first

SUMMARY

Take-aways, and how to run it

- Symmetry: DFT on 7 of 36 and 24 of 144 k-points
- SOC on demand: any spin axis and strength, no new DFT
- Spin Hall plateau e/2π at the real gap
- Altermagnets: Wannierise one spin channel, get both
- The Néel vector is just a parameter: symmetry gives the zeros, the bands give the size

RUN IT

github.com/yaroslav-zhumagulov/wannier-berri-demo

```
bash setup/setup.sh    # once, 15 min
conda activate wbdemo
bash run_all.sh        # 6 min, or the notebooks
```

Linux, macOS or WSL2, with GPAW 26.7 from conda-forge

BACKUP

Details

Kane–Mele form · nodal planes of MnTe · uniform grids · timings · references

The Wannier SOC matrix has the Kane–Mele form

$$H_{\mathrm{SO}} = i\lambda \sum_{\langle\langle ij \rangle\rangle} \nu_{ij}\, c_i^{\dagger} s_z c_j, \qquad \Delta_{\mathrm{SO}} = 6\sqrt{3}\,\lambda$$

- Second neighbours: **$\lambda \approx 3.4$ μeV**, purely imaginary, sign ±1 as in Kane–Mele
- $6\sqrt{3}\,\lambda \approx 36$ μeV: the whole gap of the full model
- $\Delta V^{x,y} = 0$: the mirror plane forbids Rashba terms, s_z is conserved
- No SOC between first neighbours: each bond lies in a vertical mirror

Shell	$\lvert d \rvert$ (Å)	$\lvert t \rvert$ (eV)	$\lvert \Delta V^{x,y} \rvert$ (eV)	$\lvert \Delta V^{z} \rvert$ (μeV)
1st, A–B	1.42	2.90	0	0
2nd, A–A	2.46	0.25	0	**3.4**
3rd, A–B	2.84	0.26	0	0
4th, A–B	3.76	0.02	0	0
5th, A–A	4.26	0.04	0	0
6th, A–A	4.92	0.03	0	0.2

Largest element per shell of $H(\mathbf{R})$ and $\Delta V(\mathbf{R})$; nothing fitted

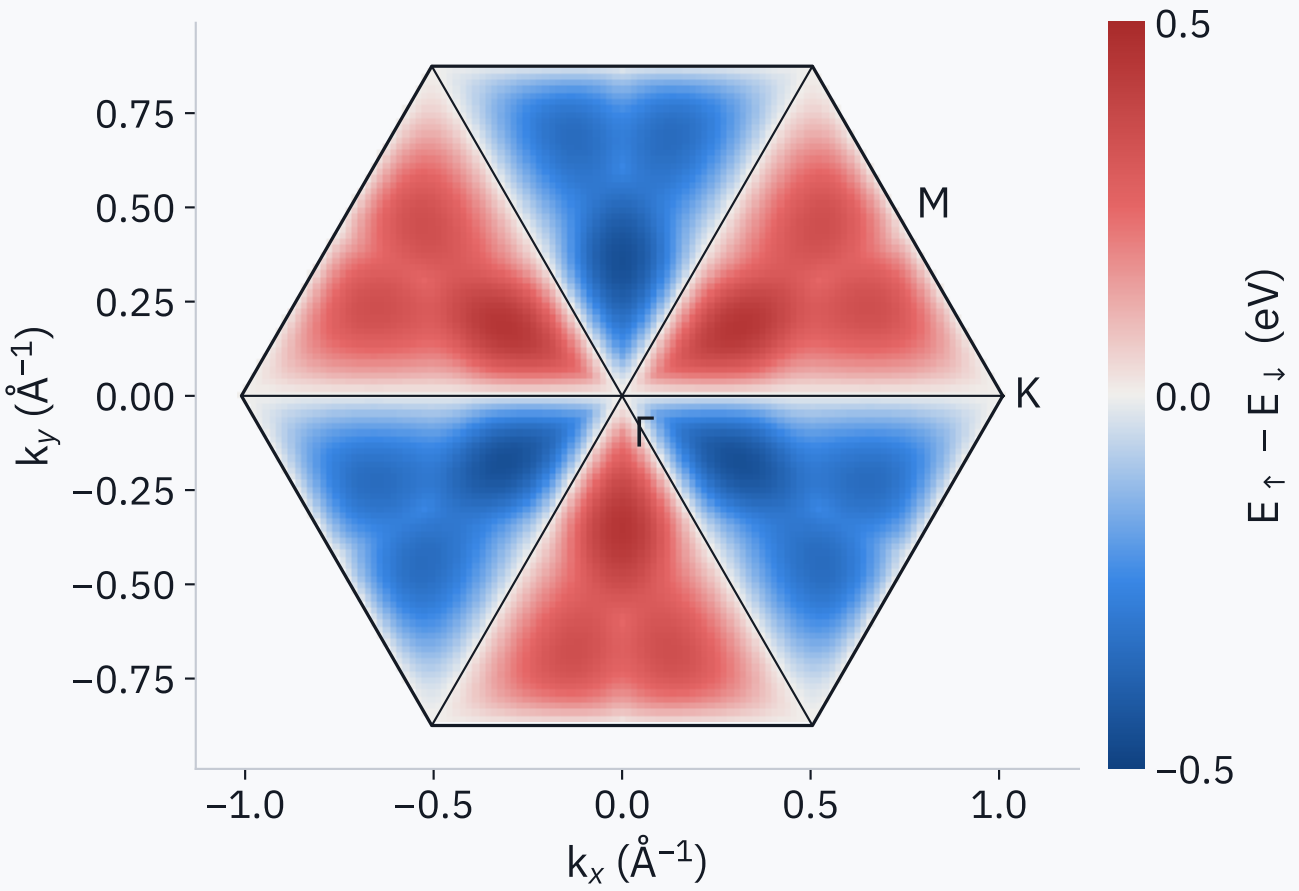
Kane, Mele, PRL 95, 226801 (2005)

Four nodal planes make MnTe g-wave

$E_\uparrow(\mathbf{k}) = E_\downarrow(R_g\mathbf{k})$ for a spin-flipping g, so the splitting vanishes on planes that R_g leaves in place:

- m_z: $k_z = 0$ and $k_z = \pi/c$
- three c-glides: three vertical planes through Γ–K

In the model the splitting is exactly zero on these planes, and up to ≈ 1 eV along Γ–M at $k_z = 0.35$ Å^{-1}



$E_\uparrow - E_\downarrow$, top valence band, $k_z = 0.35$ Å^{-1}

BACKUP

Where the time goes

Step	Graphene	MnTe
GPAW SCF + NSCF, one core	10 s	45 s
`WannierDataSOC.from_gpaw`	3.3 s	9.7 s
`wannierise`	2.1 s	9.7 s
`SystemSOC.from_wannierdata`	1.0 s	6.6 s
Conductivities (step 5)	28 s	63 s
Néel scan (step 6)	–	66 s
Animation frames	1 s	22 s
Whole notebook	1.1 min	4.2 min

Apple M2, 8 cores, 8 Ray workers: 6 minutes for everything, try-its and bonus steps included

With the Wannier model in hand, a new spin axis, SOC strength or Fermi level costs seconds

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